Bayesian Statistics I: Foundations

ECON 8310: Business Forecasting — Lecture 10

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Lecture Outline

1. Why Bayesian Thinking?
2. Bayes' Theorem
3. Prior Distributions
4. Posterior Inference with PyMC
5. Key Takeaways

Why Bayesian Thinking?

A different answer to the question of what a forecast should tell you.

Frequentist vs. Bayesian: What Is Random?

The whole difference is which thing you treat as uncertain.

	Frequentist	**Bayesian**
The parameter	A fixed, unknown constant	A quantity you are uncertain about
The data	Random — one draw of many possible samples	Fixed — it is what you observed
The answer	A point estimate and a standard error	A full distribution over values
Prior belief	No formal place for it	Stated explicitly, and updated

Everything else follows from that first row. If the parameter is fixed, you can only describe how your *estimate* would vary across hypothetical repeated samples. If the parameter is uncertain, you can describe your uncertainty about *it* directly — which is almost always the question someone actually asked.

What This Buys You in a Meeting

A 95% **confidence** interval of $[0.02,\ 0.08]$ does *not* mean a 95% chance the parameter is in it. It means the *procedure* catches the true value in 95% of repeated samples.

A 95% **credible** interval *does* mean that, given your model and data.

> One you can say to a decision-maker; the other you keep correcting.

A full distribution also answers a planner's question: *what is the chance we stock out if we order 400 units?*

The cost: you must state a prior, and defend it.

Bayes' Theorem

One equation, and it is a bookkeeping rule rather than a philosophy.

Bayes' Theorem (Gelman et al. 2013)

The update rule

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} \mid \theta)}^{\text{likelihood}} \cdot \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{evidence}}}$$

Read it as a sentence. **What I believe after seeing the data** equals **how well each value explains the data**, times **what I believed beforehand**, rescaled to sum to one.

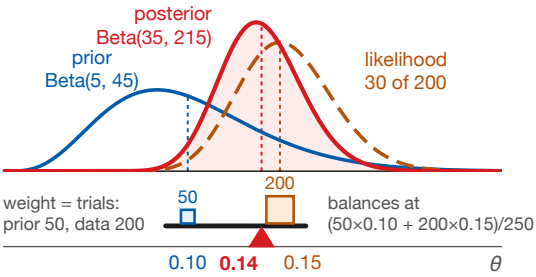
The denominator is only that rescaling — it does not change the *shape*. So the working form is:

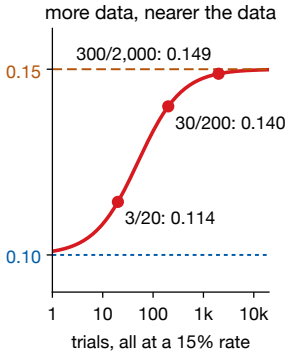
$$\text{posterior} \;\propto\; \text{likelihood} \times \text{prior}$$

You will not compute this by hand, and this course never asks you to. You are responsible for choosing the prior, checking the sampler worked, and reading the posterior.

The Update, in Pictures

A conversion rate θ: a Beta$(5, 45)$ prior, then 30 successes in 200 trials.





Multiply the curves point by point and rescale: the posterior sits between them. Its mean balances the prior's 0.10, worth 50 observations, against the data's 0.15, worth 200, at **0.14**. With little data the prior does most of the work: a feature when it is defensible, a problem when it is not.

Prior Distributions

The part you have to justify, and the part people worry about.

Priors You Will Actually Use

A prior is a distribution over a parameter, chosen to match what that parameter *can* be.

Prior	Support	Use it for
$\theta \sim \text{Beta}(\alpha, \beta)$	$[0, 1]$	Rates and proportions — conversion, click-through, defect rate
$\theta \sim \mathcal{N}(\mu_0, \sigma_0^2)$	all reals	Regression coefficients, trends, anything that can be negative
$\theta \sim \text{Exp}(\lambda)$	$[0, \infty)$	Standard deviations, waiting times, anything that must be positive

The support is the first thing to get right. Putting a Normal prior on a standard deviation allows negative values, which are impossible — the sampler will struggle and the model may not be wrong so much as incoherent.

Beta$(5, 45)$ *has mean* $5/50 = 0.1$ *— a prior centered on a 10% rate, worth about 50 observations of prior evidence.*

Choosing a Prior You Can Defend

The standard objection to Bayesian work is that the prior is arbitrary. The standard answer is that you should not be choosing arbitrary priors.

Weakly informative	Rules out the absurd, stays agnostic about the plausible. A Normal(0, 1) on a standardized coefficient says "probably not larger than about 2," which is nearly always true and nearly never controversial. **This is the default.**
Informative	Encodes real prior evidence — last year's conversion rate, a published elasticity. Defensible when you can name the source.
Flat / "uninformative"	Sounds neutral, often is not, and can be improper. Avoid it as a reflex choice.

The honest framing: a prior is an assumption stated in public, where it can be argued with. Frequentist analyses contain assumptions too — functional form, variable selection, sampling scheme — they are just not written as distributions.

Posterior Inference with PyMC

Sampling, the code, and how to tell whether the answer is trustworthy.

MCMC: The Big Idea

Beyond textbook examples, the denominator in Bayes' theorem cannot be computed. So we do not compute the posterior: we **draw samples from it**.

The picture to hold. You walk a hilly landscape in the dark, where height is posterior probability. Take a step: if the ground is higher, move; if it is lower, usually step back. Over thousands of steps you spend most of your time on the high ground.

PyMC's default sampler is NUTS, which takes smarter steps than this random walk. The intuition is unchanged.

The Samples Are the Answer

The record of where you walked *is* the answer: a histogram of the visited positions has the shape of the posterior, and nobody ever computed it.

> **Every summary is a count.** Posterior mean, credible interval, the probability of any event you care about: each is a summary of the list of visited values — counted, not derived.

The planner's stock-out question becomes one line: the share of simulated demand draws above 400 units.

PyMC: A First Model

Estimating a conversion rate — 30 successes in 200 trials

```
import pymc as pm

with pm.Model() as model:

    theta = pm.Beta("theta", alpha=5, beta=45)       # prior

    obs  = pm.Binomial("obs", n=200, p=theta,       # likelihood

                       observed=30)

    idata = pm.sample(2000, tune=1000, chains=4)     # posterior

import arviz as az

az.plot_posterior(idata)
```

A PyMC model describes **how the data could have been generated**: what I believe about the rate, and how observations arise given it. You write the story; the sampler infers. `tune=1000` discards the first 1,000 steps, while the sampler is still finding the high ground. `chains=4` runs four independent walks — which is what makes the next slide possible.

Did the Sampler Actually Work? (Martin et al. 2021)

MCMC can fail quietly. It always returns samples; whether they came from the posterior is a separate question.

Check	Want	What it means
$\hat{R}$	< 1.01	Did the chains land in the same place? If not, nothing else matters.
ESS	> 400	Effective sample size — 8,000 correlated draws may carry as much information as 200 independent ones.
Divergences	0	The sampler hit terrain it could not navigate. Even a few means the posterior is explored wrongly.

`az.summary(idata, ci_kind="hdi", ci_prob=0.94)` prints the interval, $\hat{R}$ and ESS. **Pass both arguments** — ArviZ now defaults to an 89% equal-tailed interval, not the HDI. On the model above: $\hat{R} = 1.00$, ESS 3,686, zero divergences. **Report all three.**

Prior Predictive Checks (McElreath 2020)

Before touching the data, simulate from the prior alone.

Two lines

```
with model:
    prior_pred = pm.sample_prior_predictive(500)
az.plot_ppc(prior_pred, group="prior")
```

Ask whether the simulated data could plausibly have come from your business. Impossible values — negative sales, probabilities above one — mean the prior is misspecified, not merely wide. Data spanning a billion units means it is too vague; identical simulations mean it is too tight.

A prior predictive check is **model debugging** that costs two lines — and the habit most often skipped.

Key Takeaways

Bayesian inference: beliefs in, beliefs out. Data updates your prior into a posterior.

Lecture 10: Key Takeaways

1. **Bayesian inference**: the posterior $p(\theta \mid \mathcal{D})$ is what you believe after the data.
2. **Bayes' theorem**: posterior $\propto$ likelihood $\times$ prior, landing between the two, nearer the weightier.
3. **Priors:** match the support first; weakly informative by default.
4. **MCMC** (NUTS in PyMC) draws samples; every summary is counted from them.
5. **Check before you trust**: $\hat{R}$, ESS and divergences, and a prior predictive check, every time.

Next: Bayesian Statistics II — Bayesian time series models, trend and seasonality, hierarchical models.

References I

Gelman, Andrew et al. (2013). *Bayesian Data Analysis*. 3rd. Boca Raton, FL: Chapman and Hall/CRC.

Martin, Osvaldo A., Ravin Kumar, and Junpeng Lao (2021). *Bayesian Modeling and Computation in Python*. Free online edition. Chapman and Hall/CRC. ISBN: 9780367894368. URL: `https://bayesiancomputationbook.com/`.

McElreath, Richard (2020). *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. 2nd. Boca Raton, FL: Chapman and Hall/CRC.